11.1 Resonance

WORKSHEET

The question

STEP II 1991, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The particle hangs in equilibrium with extension c. If the string has modulus λ, what is the relation between λ, a, c, m and g? What is the extension when the top has moved down by $b\sin pt$ and the particle by x?

STAGE 2 Solving

Ask yourself The equation is $\ddot{x} + n^2x = n^2b\sin pt$. Try a particular solution of the form $x = C\sin pt$. Then add the general solution of $\ddot{x} + n^2x = 0$. Which initial conditions does the particle satisfy?

STAGE 3 Staying taut

Ask yourself The string is taut exactly when its extension is positive. Write the extension $c + x - b\sin pt$ in terms of t.

Ask yourself How negative can $p \sin pt - n \sin nt$ possibly be?

Pause and try What happens to this condition as p approaches n? How small must the shaking amplitude b be when $p = 0.9n$? When $p = 0.99n$?

What's really going on

The resonance limit

Let $p \to n$ in the solution. Both the numerator and the denominator tend to zero, and a limit calculation (or l'Hôpital's rule, differentiating with respect to p) gives

$$x(t) \to \frac{b}{2}(\sin nt - nt \cos nt).$$

The term $nt \cos nt$ grows without limit: driving exactly at the natural frequency makes the amplitude increase linearly in time, each push adding to the motion in step with it. Just below resonance, the two frequencies drift in and out of step, producing *beats*: the amplitude swells and dies away with the slow frequency $n - p$ (Figure 11.1.1). The same beats are heard when two guitar strings are almost in tune.

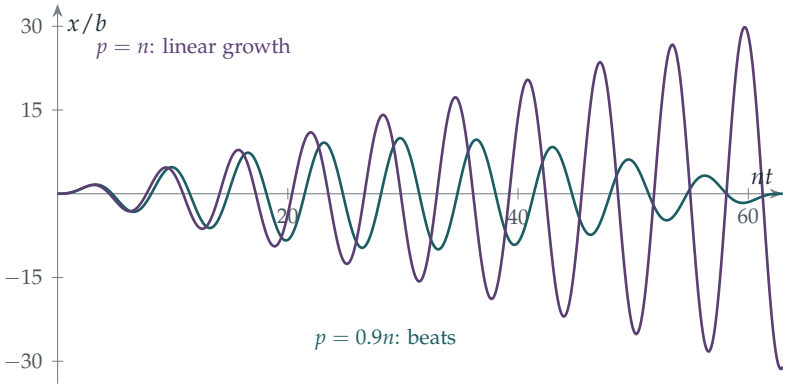


Figure 11.1.1 The response of the particle (ignoring slackness), in units of the shaking amplitude b. With $p = 0.9n$ (teal) the response swells to about ten times b and dies away again, with a beat period of $\frac{2\pi}{n-p}$. At exact resonance, $p = n$ (plum), the amplitude grows without limit.

Damping and the resonance curve

Real systems lose energy, through friction or air resistance, and that limits the growth. With damping, the steady response to a driving force at frequency p has an amplitude that peaks near the natural frequency; the sharper the peak, the less damping there is. Engineers describe this by the *quality factor* Q: roughly, the number of oscillations a system makes before losing most of its energy. A wine glass has Q in the thousands, which is why a sustained note at exactly its natural frequency can break it. A car's suspension is deliberately given a low Q, so that bumps do not set it bouncing.

Bridges that move

In 1831 a column of soldiers marching in step over the Broughton Suspension Bridge near Manchester set it vibrating until it collapsed; since then, troops have been ordered to break step on bridges. The London Millennium Bridge, opened in June 2000, began to sway sideways as soon as crowds crossed it: pedestrians unconsciously adjusted their steps to the sway, feeding it energy in step, and the bridge closed for two years while dampers were fitted. The famous collapse of the Tacoma Narrows Bridge in 1940 is often described as resonance, but it was a different phenomenon, *aeroelastic flutter*, in which the wind's force depends on the bridge's own motion. Its analysis needs the ideas of self-excited oscillation and limit cycles (Section 8.4).

Try this next

1 **A falling lift.** Try STEP II 1996, Question 11 (step.maths.org/questions/96-s2-q11): a mass on an elastic string in a lift that starts and then stops accelerating. Watch how the final amplitude depends on how long the acceleration lasts.

2 **At resonance.** Solve $\ddot{x} + n^2 x = n^2 b \sin nt$ with $x(0) = \dot{x}(0) = 0$ directly, by trying a particular solution $x = Ct\cos nt$. Check that it agrees with the limit above.

3 **Beats.** Show that $n\sin pt - p\sin nt$ can be written as a sum of two terms, one of which is approximately $2n\cos\frac{(n+p)t}{2}\sin\frac{(p-n)t}{2}$ when p is close to n. Hence explain the beat period $\frac{2\pi}{n-p}$.

4 **Damped resonance.** For $\ddot{x} + 2k\dot{x} + n^2 x = F\cos pt$ with small $k > 0$, find the steady-state amplitude, and show that it is greatest when $p^2 = n^2 - 2k^2$.

Further reading

- **A. P. French, *Vibrations and Waves*** (MIT Introductory Physics Series). Chapter 4 on forced oscillations and resonance is the classic account.
- **Steven Strogatz, *Sync*** (Penguin). How things fall into step, from fireflies to the Millennium Bridge.
- **Henry Petroski, *To Engineer Is Human*** (Vintage). Engineering failures, including the bridges in the history box.

The full guided solution is in Section 11.1:
https://beyondstep.keithmcnulty.org/ch11/resonance.html